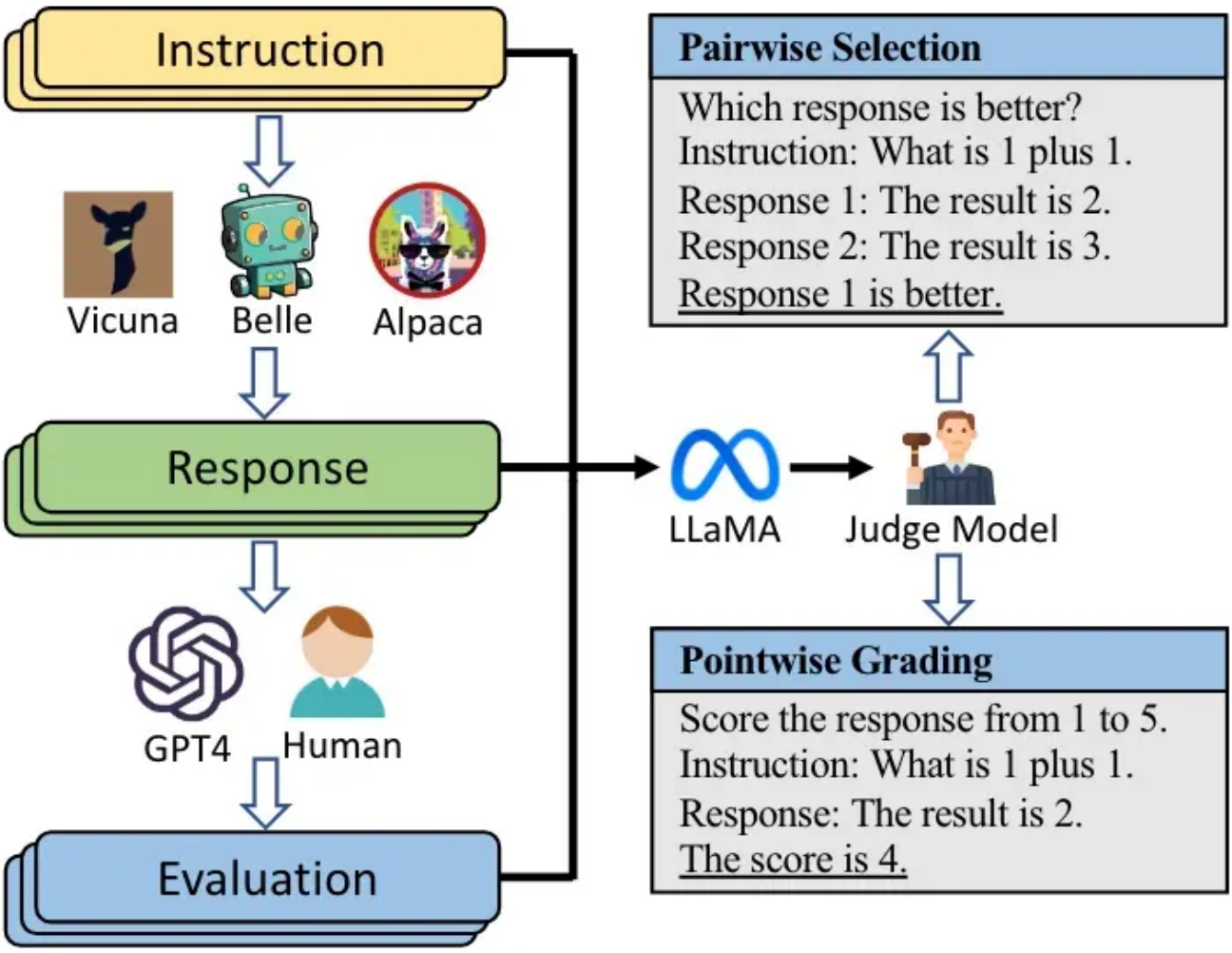

Figure 1: The general training and inference procedure of fine-tuned judge models. The training data generally comprises triplets with instruction, response and evaluation result, which are fed to the open-sourced foundation model (such as LLaMA) to create a judge model.

To assess LLM-evaluator performance on specific aspects, they used these datasets:

- LLMBar for evaluating fairness. The dataset contains paired output with a correct answer and an incorrect answer that had better superficial quality.
- HaluEval for factuality evaluation in QA, summarization, and dialogue.
- ToxiChat for toxicity evaluation based on conversations between humans and AI.
- SALAD-Bench for safety evaluations on instructions and responses.

Results: They show that the finetuned LLM-evaluators essentially functioned as task-specific

classifiers. To demonstrate this, the authors trained several LLM-evaluators, including Vicuna-generation, Vicuna-classification, and DeBERTa-classification. They found that the DeBERTa-classification evaluator performed similarly to the Vicuna models in terms of accuracy. Furthermore, these finetuned LLM-evaluators had higher correlation amongst themselves than with gpt-4. Taken together with further findings below, this suggests that the finetuned LLM-evaluators were inherently task-specific classifiers.

Model	JudgeLM-test		PandaLM-test		Auto-J-test	Prometheus-test	
	accuracy	F1	accuracy	F1	agreement	pearson-ind	pearson-ood
Released Models†	78.98	68.62	68.97	60.95	54.6	0.864	0.869
Vicuna-generation‡	**82.44**	**71.77**	**72.37**	**60.78**	**47.6**	0.826	0.815
Vicuna-classification‡	82.16	70.07	70.87	60.34	46.8	**0.846**	**0.831**
DeBERTa-classification‡	81.30	68.34	72.27	51.75	31.7	0.835	0.813
GPT-3.5-0613	72.57	51.40	64.36	46.40	42.7	0.636	0.563
GPT-4-1106-preview	85.28	76.87	74.07	68.09	56.3	0.742	0.743

Table 8: Comparison of generation and classification-based evaluators. Results with † are from evaluating the four publicly released models on their respective testsets, and results with ‡ are from evaluating models trained by us.

Task	Perturbation Axis	Description
LF	GRAMMAR	Introducing grammatical errors in the answer. Eg: This is good → This are good.
	SPELLING	Introducing "valid" spelling errors in the answer. Eg: Toxicity → Tocixity.
	CONSISTENCY	Introducing errors in the "consistency" of the answer (like tone, terminology, etc.)
	CHRONOLOGY	Introducing errors in the chronological or the logical flow of the answer.
	COHERENCE	Introducing errors that affect the coherence of the answer.
	COMPREHENSIVENESS	Introducing vagueness, irrelevance or lack of context in the answer.
F	CONTEXTUAL	Replacing fact with a contextually similar incorrect fact. Eg: electricity → magnetism.
	ENTITY	Replacing a named entity with an incorrect entity. Eg: Poland → London.
	INCORRECT FACT	Adding a new contextually relevant incorrect fact in the answer.
	NUMBER ERRORS	Introducing errors in the various numbers reported in the answer. Eg: 1987 → 1887.
	OPPOSITE FACT	Replacing a fact in the answer with its negation. Eg: ... will have ... → ... wont have ....
	REMOVE FACT	Removing a fact critical to the correctness and completeness of the answer.
IF	DO LESS	Doing less than what is *explicitly* requested in the question.
	DO MORE	Doing more than what is *explicitly* requested in the question.
	IGNORE FORMAT	Ignoring the formatting and other constraints mentioned in the question.
	SEQUENCE ERRORS	Ignoring the sequence in the response when *explicitly* requested in the instruction.
	ASSUMPTIONS	Making new incorrect assumptions about the instruction.
R	CALCULATIONS	Introducing calculation errors in the answer. Eg: $2+3=5 \rightarrow 2+3=6$
	COPYING NUMBERS	Introducing errors while considering the numbers mentioned in the instruction.
	FINAL ERRORS	Introducing errors only the final reported answer while retaining the correct solution.
	INCORRECT UNITS	Introducing errors in the units reported and considered in the answer.
	WRONG FORMULA	Introducing errors in the formula used in the answer. Eg: $\pi r^2 \rightarrow 2\pi r$
SI	SCORE INVARIANT	Introducing modifications in the answer which would not result in a score penalty.

Table 2: Perturbation categories across each of the task abilities. The green highlights indicate the original text and the red highlights indicated the perturbed text. Complete examples of each perturbation can be found in supplementary material.

[LLMs instead of Human Judges: A Large Scale Empirical Study across 20 NLP Evaluation Tasks](#)evaluates 11 LLM-evaluators to replicate human judgment across 20 language tasks. These include general tasks such as reasoning, instruction following, and toxicity detection, as well as downstream tasks such as summarization, translation, and dialogue.

Dataset	Task	Size	Type	Guidelines	Expert	Agreement	Leaked
CoLA (Warstadt et al., 2019)	Acceptability	1,043	Categorical	✗	✓	✓	✓
CoLA grammar (Warstadt and Bowman, 2020)	Acceptability	1,043	Categorical	✗	✓	✗	✓
Switchboard (Wallbridge et al., 2022)	Acceptability	100	Graded	✓	✗	✗	
Dailydialog (Wallbridge et al., 2022)	Acceptability	100	Graded	✓	✗	✗	
Inferential strategies (Mondorf and Plank, 2024)	Reasoning	300	Categorical	✓	✓	✗	✗
ROSCOE (Golovneva et al., 2023)	Reasoning	756	Categorical + Graded	✓	✓	✗	
Recipe-generation (Stein et al., 2023)	Planning	52	Graded	✓		✗	
Medical-safety (Abercrombie and Rieser, 2022)	Toxicity & Safety	3,701	Preference	✓	✓		
DICES (Aroyo et al., 2023)	Toxicity & Safety	1,340	Categorical	✗	Mixed	✓	
ToxicChat (Lin et al., 2023)	Toxicity & Safety	5,654	Categorical	✗	✓	✗	
Topical Chat (Mehri and Eskenazi, 2020)	Dialogue	60	Graded + Categorical	✗	✓	✓	
Persona Chat (Mehri and Eskenazi, 2020)	Dialogue	60	Graded + Categorical	✗	✓	✓	
WMT 20 EnDe (Freitag et al., 2021)	Machine Translation	14,122	Graded	✗	✓	✓	
WMT 20 ZhEn (Freitag et al., 2021)	Machine Translation	19,974	Graded	✗	✓	✓	
WMT 23 EnDe (Kocmi et al., 2023)	Machine Translation	6,588	Graded	✗	✓	✗	
WMT 23 ZhEn (Kocmi et al., 2023)	Machine Translation	13,245	Graded	✗	✓	✗	
G-Eval / SummEval (Liu et al., 2023)	Summarisation	1,600	Graded	✓		✓	✓
QAGS (Wang et al., 2020)	Summarisation	953	Categorical	✓	✗	✗	
NewsRoom (Grusky et al., 2018)	Summarisation	420	Graded	✓	✗	✗	✓
LLMBar (Zeng et al., 2024)	Instruction Following	419	Categorical	✓	✓	✓	✗

Table 2: Overview of the main features of the datasets considered in the study. Note that 'Size' refers to the number of annotated samples, not to the total number of human annotations. 'Agreement' indicates whether multiple annotations were available for the same instance or not. Information on possible data leakage was retrieved from Balloccu et al. (2024).

The authors selected 11 widely used models that had high performance across several tasks on the Open LLM and Chatbot Arena leaderboards. These include gpt-4, gemini-1.5, command-r, command-r+, llama-3-8b, llama-3-70b, mistral, mixtral-8x7b, mixtral-8x22b, olmo, and starling.
Results: The LLM-evaluators had high variance in correlation with human judgments across the datasets. Each model performed poorly on some datasets, suggesting that they're not reliable enough to systematically replace human judgments.

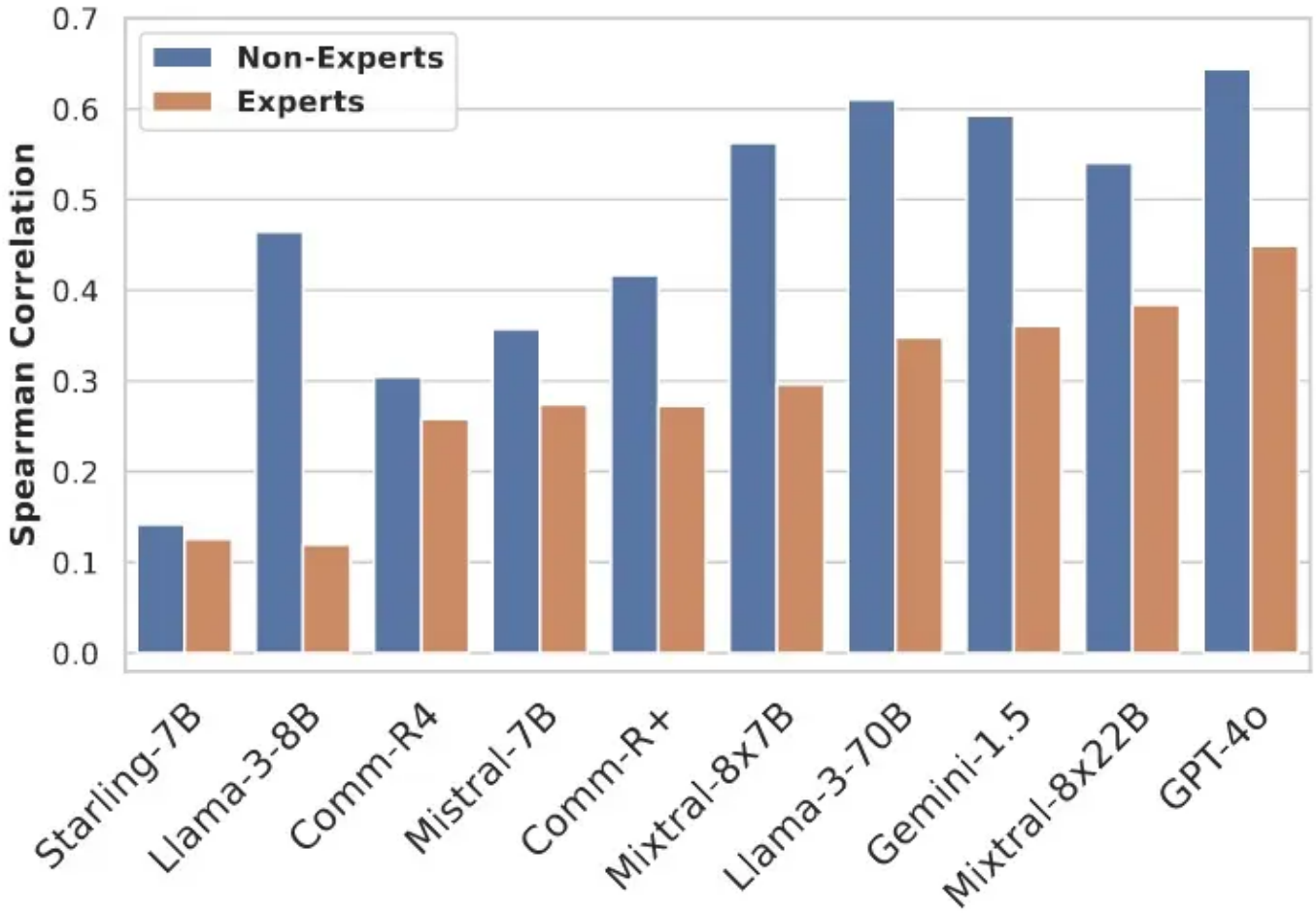

Figure 2: Model correlation with human experts vs. non-expert judgments (OLMo has a negative correlation and does not appear in the figure).

[udging the Judges: Evaluating Alignment and Vulnerabilities in LLMs-as-Judges](#) evaluates nine LLM-evaluators, using the TriviaQA dataset as a knowledge benchmark. The researchers sampled 400 questions from the unfiltered partition of TriviaQA and used the short answers as reference answers (i.e., the evaluation approach is reference-based). The training set was used as few-shot examples.

The LLM-evaluators were instructed to only respond with a single word: “correct” or “incorrect”. As baselines, the authors included exact match (EM) and contains substring (contains). For alignment metrics, they considered percentage agreement and Cohen’s .

```
Your task is to look at the following question, and based on
the references provided,
```

```
determine if the model’s response is correct or incorrect.
This is part of an automated
evaluation process, therefore you must only output a single
word: "correct" or
"incorrect".

Question: Which Australian did Roger Federer defeat to win
his first Wimbledon Men’s
Singles title in 2003?

References:
MARK PHILIPPOUSSIS
MARK PHILIPPOUSSIS

Model Response:
Mark Philippoussis

Evaluation (correct/incorrect):
```

Results: Gpt-4 and llama-3-70b had good human-alignment, achieving Cohen’s of 0.84 and 0.79 respectively. However, they were still significantly lower than the human-human Cohen’s of 0.97. Surprisingly, the contains baseline had higher correlation than half of the evaluator models on what was essentially a fuzzy matching task.

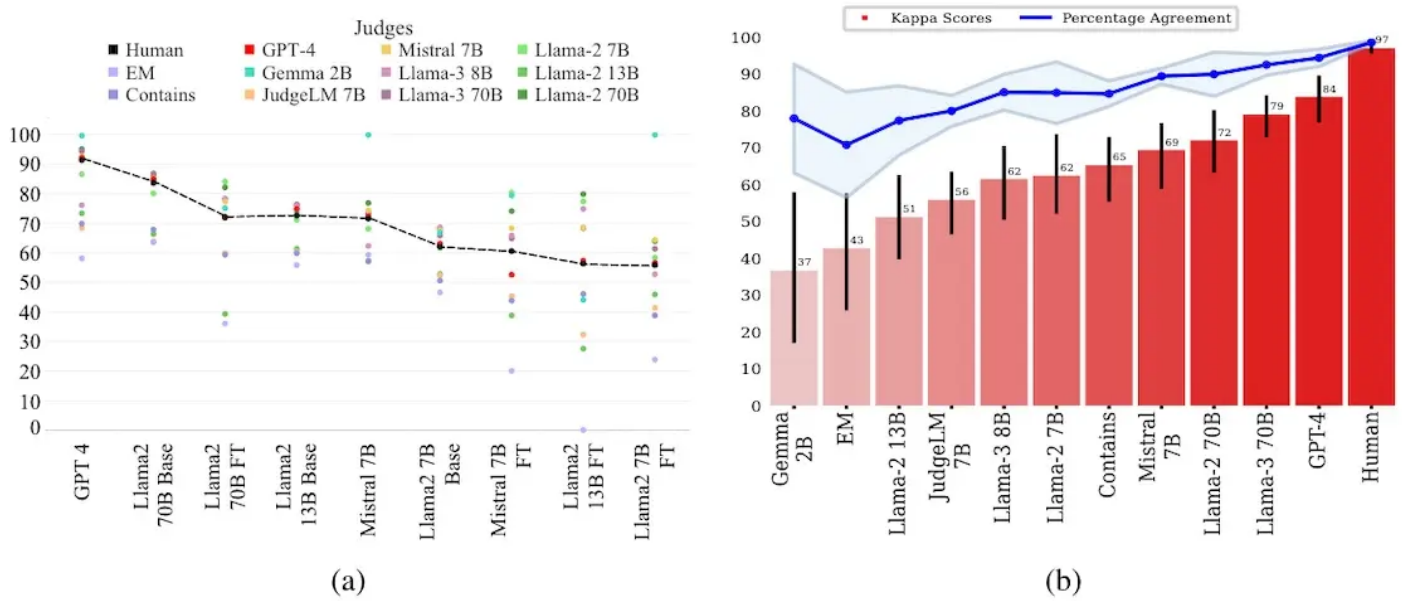

Figure 1: (a) Scores assigned to all exam-taker models by the various judge models. (b) Average percent agreement (blue line) and Kappa scores (red bars) of judge models with human judges. Error bars annotate standard deviation across exam-taker models. Alignment is poor for most judge models, but both `Llama3 70B` and `GPT-4 Turbo` have Cohen's Kappa coefficient that are indicative of excellent alignment (79 and 84, respectively), though still well below the human alignment score of 96.